Multi-Agent Reinforcement Learning for Connected and Automated Vehicles Control: Recent Advancements and Future Prospects

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***Abstract*—Connected and automated vehicles (CAVs) have emerged as a potential solution to the future challenges of developing safe, efficient, and eco-friendly transportation systems. However, CAV control presents significant challenges significant challenges due to the complexity of interconnectivity and coordination required among vehicles. Multi-agent reinforcement learning (MARL), which has shown notable advancements in addressing complex problems in autonomous driving, robotics, and human-vehicle interaction, emerges as a promising tool to enhance CAV capabilities. Despite its potential, there is a notable absence of current reviews on mainstream MARL algorithms for CAVs. To fill this gap, this paper offers a comprehensive review of MARL's application in CAV control. The paper begins with an introduction to MARL, explaining its unique advantages in handling complex and multi-agent scenarios. It then presents a detailed survey of MARL applications across various control dimensions for CAVs, including critical scenarios such as platooning control, lane-changing, and unsignalized intersections. Additionally, the paper reviews prominent simulation platforms essential for developing and testing MARL algorithms for CAVs. Lastly, it examines the current challenges in deploying MARL for CAV control, including safety, communication, mixed traffic, and sim-to-real challenges. Potential solutions discussed include hierarchical MARL, decentralized MARL, adaptive interactions, and offline MARL. The work has been summarized in MARL_in_CAV_Control_Repository.**

***Note to Practitioners*—This paper explores the application of MARL for controlling CAVs in complex traffic scenarios such as platooning, lane-changing, and intersections. MARL provides an adaptive and decentralized control approach, offering advantages over traditional rule-based and optimization-based methods in handling dynamic interactions, improving traffic flow, enhancing safety, and optimizing fuel efficiency. The paper reviews state-of-the-art MARL algorithms and simulation platforms, serving as a resource for practitioners looking to implement these advanced control strategies. However, real-world deployment remains challenging due to communication reliability, real-time decision-making constraints, and the complexities of mixed traffic environments involving both automated and human-driven vehicles. Future research should focus on ensuring robust inter-agent communication, developing safety-aware MARL frameworks, and addressing the sim-to-real transfer gap. This paper provides insights to help practitioners bridge the gap between research and deployment, facilitating the development of more scalable, adaptive, and reliable CAV control.**

***Index Terms*—Connected and automated vehicles, multi-agent reinforcement learning, intelligent transportation systems, vehicle control.**

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I. Introduction

ADVANCEMENTS in automation, artificial intelligence (AI), and vehicular communication are transforming transportation, leading to the emergence of connected and automated vehicles (CAVs) [1]. By integrating vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, CAVs enable cooperative perception, real-time decision-making, and traffic optimization [2]. However, achieving safe, efficient, and scalable CAV control in complex and dynamic environments remains a critical challenge [3], [4].

Various control optimization strategies have been explored to enhance the coordination and efficiency of CAVs [5], [6]. Among these, conventional methods such as model predictive control (MPC) have been employed to coordinate platoon behavior, minimize control delay, and reduce traffic oscillations [7]. Additionally, non-convex optimal control problems, such as CAV coordination at intersections, have been tackled with semidefinite relaxation techniques [8]. However, these approaches often depend on precise system modeling, which is not always available [9], and require substantial computational resources, making them impractical for real-time control [10].

Besides, reinforcement learning (RL) has gained increasing attention within the research field due to its outstanding abilities in addressing sequential decision-making tasks, such as gaming [11], robotics [12], behavioral planning [13], intelligent energy management [14], [15]. Similarly, AV control has begun exploring the potential of employing RL for various

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traffic scenarios [16]. For instance, in [17], a deep deterministic policy gradient (DDPG)-based model for platooning control is introduced to enhance fuel economy, driving efficiency, and safety at signalized intersections through real-time optimization. This approach featuring an effective reward function, demonstrates strong performance under varying traffic demands and traffic light cycles with different durations. Taking into account the role of HDVs in the control of AVs, Shi et al. employ a distributed proximal policy optimization (DPPO) control strategy that allows them to learn from and respond to the behavior of the HDVs, optimizing performance at both the local subsystem level and the broader mixed traffic context [18]. Qu et al. introduce a control approach utilizing DDPG algorithm aimed at reducing traffic fluctuations and enhancing fuel economy. However, these RL-based approaches only consider a single agent (i.e., AV), making them poorly suited for multi-agent coordination and generalization in dense traffic environments [19]. As autonomous driving moves toward multi-vehicle interaction and cooperative control, single-agent RL struggles with scalability, decentralized decision-making, and shared state information, limiting its effectiveness in complex, dynamic traffic scenarios.

Multi-agent reinforcement learning (MARL) extends single-agent RL to settings where multiple agents interact and learn collectively. This innovative approach has found applications across a broad spectrum of fields. Notably, the gaming and simulation industry has utilized MARL to create more complex and interactive environments [20]. Furthermore, within the realm of finance, MARL models are employed to simulate the dynamics of markets and the behaviors of agents [21]. In traffic control and management, MARL algorithms have been instrumental in optimizing traffic flows and reducing congestion. The realm of AVs and robotics has also benefited greatly, with MARL enabling cooperative navigation and coordination among vehicles and facilitating collaborative tasks in AVs and robotics [22]. Extended to CAV control, each vehicle acts as an independent agent that learns from its interactions with the surrounding environment, which includes other vehicles, traffic signals, and road conditions. Each CAV learns not only to maneuver safely and efficiently on its own but also to cooperate and communicate with other CAVs [23]. Therefore, MARL offers a valuable strategy for improving the performance and efficiency of CAVs.

Several studies have empirically demonstrated that MARL outperforms single-agent RL in scenarios requiring multi-agent interaction, such as mixed traffic and cooperative vehicle control [24]. Unlike RL, which follows an independent decision-making framework, MARL enables policy coordination and shared decision-making, resulting in superior scalability, efficiency, and safety in dense and dynamic environments [25]. While [26] provides an overview of single-agent RL in CAV control, it highlights the limitations of independent decision-making in multi-agent environments, where interactions between vehicles significantly impact performance. These limitations underscore the necessity of MARL for cooperative lane-changing, intersection management, and on-ramps merging optimization. In [27], the authors emphasize that existing safe RL methods designed for single-agent RL cannot be directly applied to MARL scenarios, reinforcing that RL struggles to handle multi-agent coordination challenges in safety-critical CAV applications. Similarly, in [28], MARL-based motion controllers are evaluated against RL approaches for lane-changing, showing that MARL achieves smoother lane changes, lower collision rates, and better adaptability to dynamic environments. These findings confirm that MARL outperforms RL when multiple autonomous agents need to be coordinated in an interactive traffic environment. While MARL offers clear advantages, it also introduces additional computational and communication complexity, particularly in terms of information sharing during training and execution. A common concern is that MARL may require excessive connectivity and centralized training, making it impractical for real-world deployment. To mitigate these challenges, recent works have explored hierarchical MARL [18], graph-based coordination [20], and decentralized training techniques [29], ensuring scalability without excessive information-sharing demands.

In the context of CAV control, MARL has been increasingly adopted to optimize collective behaviors, offering a scalable and cooperative solution to real-world deployment challenges [30]. Connectivity within this cooperative deployment operates on two levels: V2V communication, which enables direct data sharing between vehicles, and V2I communication, which expands connectivity through technologies like vehicular ad hoc networks and Ethernet-based systems. Together, these communication channels ensure robust inter-agent communication and situational awareness, both of which are critical for the scalability and effectiveness of CAVs [31]. Fig. 1 illustrates a comprehensive multi-agent system, i.e., CAVs, including environmental data from sensors such as GPS/INS, cameras, radar, and lidar, along with V2I communication channels providing information from traffic signals, roadside units, and cloud systems. This data is fused to create a unified environmental representation. Each AV maintains a local data repository while accessing shared V2V and V2I information, enhancing decision-making and situational awareness [32]. The decision-making and motion-planning modules process these inputs to generate optimal strategies for safe and efficient navigation. These strategies are then executed through steering and powertrain control modules. By leveraging both V2V and V2I communication, the multi-agent system demonstrates how integrated connectivity enables robust coordination and scalability within the CAV ecosystem, ultimately laying the foundation for smarter and more efficient transportation networks.

Despite advancements in MARL for CAV systems, significant challenges remain in adapting these techniques to real-world CAV control. Issues such as scalability, robustness in dynamic environments, and the integration of decentralized learning frameworks require further investigation. While a few articles describe the potential of MARL in CAVs [33], none of them, to the best of our knowledge, are specifically devoted to the application of MARL for CAV control. This paper is thus intended to deliver a comprehensive and systematic review of MARL within the realm of CAVs, such as lane changing, platooning control, traffic signal cooperation, and on-ramp

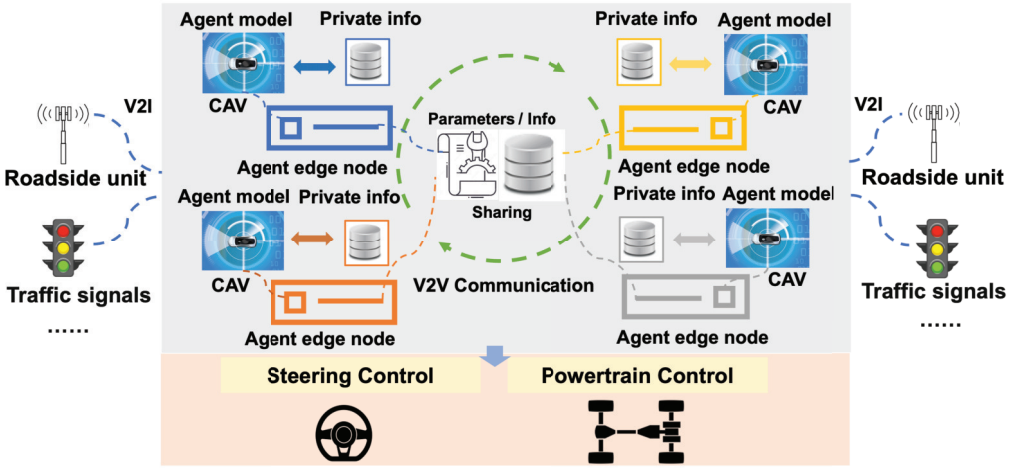

Fig. 1. The schematic of the multi-agent system for CAV control.

merging. Through this review, we not only provide a clear, up-to-date understanding of the current classic MARL algorithms applied in CAV control, but we also outline the potential direction for future work in this area. The main contributions of the work presented in this paper are summarized as follows:

1) We survey the recent developments in MARL algorithms, discussing their diverse applications in aspects of CAV control based on the extent of control dimensions. Furthermore, we provide an extensive examination of the leading simulation platforms employed in MARL research for CAV control.
2) We highlight and explore the technical challenges that MARL faces in these applications and discuss potential research directions to address these challenges.
3) The work, including a well-classified list of relevant papers, has been presented at the following site: https://github.com/huahuaedi/MARL_in_CAV_control_review.

This paper is organized as follows: Section II introduces the basics of RL and MARL; A detailed review of MARL applied in CAV control from different control dimensions and mainstream simulation platforms are described in section III. Section IV summarizes the remaining challenges and opportunities. Finally, Section V concludes the review.

II. Methodologies

In this section, we begin with a comprehensive overview of the fundamental principles of RL. We then delve into various prominent MARL algorithms, establishing the context to enhance the understanding of our review.

A. Preliminaries of RL and MARL

RL, typically formalized as a Markov Decision Process (MDP), provides a framework for sequential decision-making under uncertainty [34]. An agent interacts with the environment by taking actions and receiving rewards, aiming to learn an optimal policy that maximizes cumulative rewards. The MDP is defined as $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $\mathcal{P}$ the transition probability, and $\mathcal{R}$ the reward function.

Deep RL methods leverage neural networks to approximate value functions or policies, enabling applications in high-dimensional control problems such as autonomous driving. These methods can be broadly categorized into: **Value-based methods** (e.g., DQN, DDQN): learn the Q-function $Q(s, a)$ to select actions via maximization; **Policy-based methods** (e.g., DDPG): optimize a parameterized policy π_θ directly using gradient ascent; **Actor-critic methods** (e.g., Advantage Actor-Critic (A2C) [35]): combine policy and value function learning to improve stability and reduce variance.

In multi-agent settings such as CAV control, MARL extends the RL paradigm to systems where multiple agents interact and jointly influence outcomes [36]. Agents are modeled as nodes in a communication graph $\mathcal{G} = (\nu, \varepsilon)$ and operate under decentralized and partially observable conditions, typically formalized as a partially observable MDP (POMDP).

B. Training and Execution Strategies in MARL

In MARL settings, different approaches and frameworks are employed to tackle the complex challenges of coordinating and training agents in decentralized environments. This subsection explores two fundamental paradigms: Centralized Training with Decentralized Execution (CTDE) and Decentralized Training with Decentralized Execution (DTDE), each offering distinct insights and trade-offs in the pursuit of effective multi-agent learning [37], [38], [39].

1) Centralized Training With Decentralized Execution (CTDE): In this framework, the decentralized problem is initially transformed into a centralized one, solvable by a central controller, which gathers essential training information, e.g., observation, reward, and global state information, for the training process. Following this data collection, the centralized value functions are learned based on the information of all agents, and then the gradient from the centralized value function is used to train the policy of each agent. But in the process of execution, each agent outputs action according to its individual observation (see Fig. 2).

It is important to acknowledge the inherent trade-offs of the CTDE. One significant strength is the enhanced learning

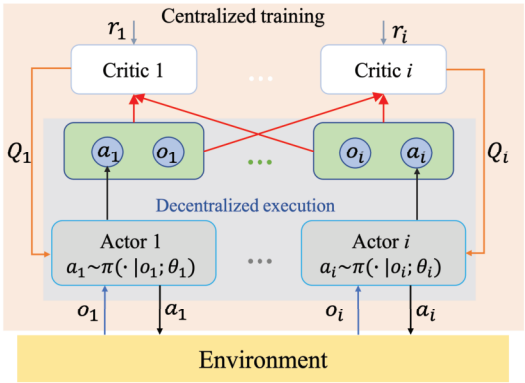

Fig. 2. Illustration of centralized training with decentralized execution (CTDE) in MARL.

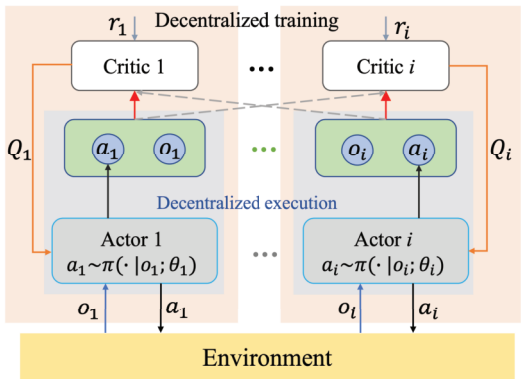

Fig. 3. Illustration of decentralized training with decentralized execution (DTDE) in MARL.

stability enabled by centralized training, which can result in more robust policies [40]. However, it is crucial to recognize that maintaining a centralized controller comes with its own set of challenges. Firstly, it can be prohibitively expensive and infeasible in certain scenarios, particularly in large-scale or resource-constrained environments. For example, in large-scale CAV networks such as urban traffic control [41] or highway platooning [42], maintaining a centralized critic requires high bandwidth and computational resources, making real-time learning impractical. Similarly, resource constraints in edge-computing environments limit the feasibility of processing global information centrally, favoring decentralized approaches [43]. Additionally, the centralized controller introduces privacy concerns, as it necessitates the sharing of sensitive information among agents. Moreover, the central controller becomes a single point of failure, rendering the whole system vulnerable to disruptions. These trade-offs highlight the need for careful consideration of the CTDE framework's suitability in specific MARL applications, taking into account the advantages and drawbacks it presents.

2) Decentralized Training With Decentralized Execution (DTDE): In DTDE, each agent learns and acts independently, relying solely on local observations and policy updates without a central controller. Unlike centralized or partially centralized frameworks, DTDE agents do not require global state information but instead learn through trial-and-error interactions with the environment. A key challenge in DTDE is maintaining learning stability due to non-stationarity, as agents' policies evolve independently over time. Traditional independent Q-learning (IQL) [44] and independent advantage actor-critic (IA2C) [45] are widely used but often struggle with convergence due to the lack of inter-agent coordination [46]. To mitigate non-stationarity, Zhang et al. proposed a fully decentralized MARL framework, where agents exchange messages with neighboring agents, improving coordination and offering theoretical convergence guarantees [47]. Fig. 3 illustrates the DTDE framework based on the actor-critic architecture, where each agent independently updates its policy and critic function based on local interactions with the environment.

3) Offline Reinforcement Learning: Beyond CTDE and DTDE, Offline RL provides an alternative learning paradigm that relies exclusively on pre-collected datasets, eliminating the need for continuous environment interaction during training. Unlike traditional online RL, which explores and updates policies dynamically, Offline RL learns solely from fixed datasets, making it particularly useful in scenarios where real-time exploration is expensive, unsafe, or impractical [48]. However, a major challenge in Offline RL is the issue of distributional shift, when the learned policy encounters states or actions that are insufficiently represented in the dataset, leading to suboptimal or unsafe decision-making. To address this, various techniques, such as policy regularization [49], conservative Q-learning [50], and behavior cloning [51], have been proposed to improve policy robustness and generalization. Given its potential for leveraging large-scale offline data, Offline RL continues to be an active research area with broad applications across different domains.

C. MARL Algorithm Variants

In this subsection, we have exclusively reviewed the common MARL algorithms that find utility in CAV applications. Further, the MARL algorithms are divided into four categories: value function decomposition, learning to communicate, hierarchical structure, and causal inference, which are rooted in the inherent complexity and diverse requirements of multi-agent environments, particularly in CAV control. Table I offers an extensive overview of key works from four distinct perspectives within various settings.

1) Value Function Decomposition: The challenge of credit assignment in cooperative settings has emerged as a significant area of research interest. The shared rewards in a fully cooperative environment make it difficult to distinguish the contribution of each agent, some agents tend to be lazy [72]. To solve the above problem, some studies learn different value functions to distinguish the contribution of each agent. Foerster et al. [73] and Guo et al. [74] introduced the counterfactual baseline principle to learn different value functions by centralized learning method in the context of a shared team reward environment. Additionally, Hou et al. have developed a credit allocation mechanism based on role attention, which facilitates the learning process of role policies in multi-agent systems by managing how credit is allocated among agents [75]. Liu et al. construct the causal effect of the agent's actions on the

TABLE I
AN OVERVIEW OF THE PRIMARY WORKS FOR MARL ALGORITHMS

Categories	Work	MARL algorithm	Novelty
Value Function Decomposition	[52], 2020	WQMIX	Introducing weighting functions and non-monotonic mixing networks
	[53], 2021	AVD-Net	An attention-based approach to capitalize on the coordination relations between agents.
	[54], 2021	RADAR	A resilient adversarial approach to enhance robustness against agent failures or behavioral changes
	[55], 2022	VGN	Utilizing graph attention networks to solve value functions, improving coordination in cooperative tasks
	[56], 2023	PER	Both the reward of an agent acting on its own and cooperating with other observable agents
	[57], 2024	MA-MIX	A multi-head attention mechanism to enhance information exchange and cooperation among agents
Learning to Communicate	[58], 2020	NeurComm	Encoding and concatenating communicating signals rather than aggregating them
	[59], 2020	IMAC	Learning an efficient communication protocol based on the information bottleneck principle
	[60], 2024	Verco	Embedding large language models into agents to generate human-understandable verbal communication
	[61], 2025	AsynCoMARL	Leveraging graph transformers to learn communication protocols from dynamic graphs
Hierarchical Structure	[62], 2019	HSD	Training a cooperative decentralized policy for high-level skill selection
	[63], 2019	RODE	Introducing a method to clearly divide the action space through action clustering
	[64], 2022	H-MARL-PQ	A hierarchical architecture that methodologically combines multi-agent with single-agent RL
	[65], 2023	HAVEN	Designing a dual coordination mechanism within a two-level hierarchical structure
	[66], 2023	HMASD	A transformer structure for sequential skill assignment and low-level policy learns for individual skills
Causal Inference	[67], 2022	FD-MARL	Constructing continuous communication protocols based on causal analysis
	[68], 2022	DVD	Deconfounded value function decomposition based on causal effects
	[69], 2023	ICL	Evaluating the causal effect of each agent's observations on team collaboration performance
	[70], 2023	LAIES	Calculating the causal effect of their actions on external states using the do-calculus process
	[71], 2024	DMS	Individual intention inference and causal reasoning to enable fully decentralized coordination

external state from the perspective of a causal diagram, solving the problem of lazy agents [70].

Among the above methods to solve credit assignment, value function decomposition has remained a key approach for addressing credit assignment challenges in cooperative MARL by decomposing the joint value function into individual components. Building upon foundational methods like VDN [76] and QMIX [77], several advanced algorithms have been developed to tackle increasingly complex scenarios. For instance, the Value Decomposition with Graph Attention Networks (VGN) method employs graph attention networks to dynamically model agent relationships, improving coordination in cooperative tasks [55]. Similarly, the MA-MIX algorithm integrates a multi-head attention mechanism to enhance information exchange and cooperation among agents [57]. These advancements underscore the ongoing evolution of value function decomposition techniques, further improving agent coordination, communication efficiency, and adaptability in increasingly complex environments.

2) Learning to Communicate: Research in MARL can be categorized into two distinct groups according to their methods of communication [58]. The first group operates without communication among agents and centers its efforts on stabilizing training by employing advanced value estimation methods. For example, MADDPG [78] extends DDPG to the MARL setting by employing a centralized critic network to enable the global value estimate. Similarly, COMA [73] modifies the actor-critic method for use in MARL, calculating the advantage function for each agent by employing a centralized critic in conjunction with a counterfactual baseline. The second group investigates explicit communication mechanisms, which can be predefined heuristic protocols or learnable communication strategies. FingerPrint [79] stabilizes learning under non-stationary policies by sharing low-dimensional policy fingerprints among agents. DIAL [80] enables differentiable communication by having DQN-based agents simultaneously generate and process messages via backpropagation. Further advancements have sought to optimize communication efficiency and reduce redundant message exchange. NeurComm [58] proposes that encoding and concatenating communication signals, rather than simply aggregating them, effectively mitigates information loss and enhances coordination among agents. This approach allows for structured message processing, improving decision-making in dynamic environments. Similarly, IMAC [59] formulates an efficient communication protocol based on the information bottleneck principle, ensuring that agents only transmit the most relevant information to their peers. The framework develops a differentiable communication strategy, incorporating state, policy, and encoded neighboring information, which is then concatenated into the state space of nearby agents.

More recent works have further expanded communication capabilities to improve interpretability, adaptivity, and robustness in MARL. Verco [60] integrates large language models (LLMs) into MARL agents, enabling human-readable verbal communication that enhances coordination and transparency. Meanwhile, AsynCoMARL [61] introduces graph transformer-based learning to dynamically adjust communication protocols in real time. By leveraging dynamic graph structures, agents can learn when and how to communicate, ensuring scalable and adaptive multi-agent cooperation. These developments mark a shift towards structured, efficient, and learnable communication mechanisms. Instead of relying on predefined message-passing rules, modern approaches prioritize adaptive protocols, minimal yet effective communication, and scalable coordination strategies, improving MARL performance in complex, multi-agent environments.

3) Hierarchical Structure: In MARL, hierarchical structures play a crucial role in addressing challenges such as sparse rewards, complex decision-making, and limited transferability. Hierarchical MARL enables agents to learn decision-making strategies across different levels of temporal abstraction, making it particularly beneficial for large-scale coordination tasks. Existing hierarchical MARL methods can be broadly categorized into option-based hierarchies [81], [82] and goal-based hierarchies [83], [84], both of which have been successfully applied to multi-agent cooperation.

Several hierarchical MARL frameworks have been introduced to enhance scalability, coordination, and efficiency. Hierarchical Skill Discovery (HSD) trains decentralized policies for high-level skill selection while simultaneously learning low-level execution policies, improving adaptability in dynamic environments [62]. Role-based Decomposition (RODE) further refines action space partitioning through action clustering, ensuring that each action corresponds to a distinct subspace, thereby reducing the need for extensive parameter tuning [63]. Hierarchical Multi-Agent Reinforcement Learning with PER-QMIX (H-MARL-PQ) combines multi-agent and single-agent RL through a hierarchical framework, integrating macro-operations and prioritized experience replay to accelerate learning in complex wargames with sparse rewards [64]. To further improve hierarchical coordination, HierArchical Value dEcompositioN (HAVEN) introduces a dual-coordination mechanism that operates both within hierarchical layers and across agents, optimizing strategy execution at multiple levels and enhancing policy stability [65]. Meanwhile, Hierarchical Multi-Agent Skill Discovery (HMASD) employs a transformer-based structure for sequential skill assignment, allowing low-level policies to autonomously discover and refine agent-specific skills, enhancing flexibility in cooperative decision-making [66].

Despite these advancements, hierarchical MARL faces key challenges in designing dynamic hierarchical structures. Unlike static hierarchies with predefined roles, dynamic structures evolve based on task complexity, agent interactions, and environmental changes, enabling adaptive role assignment, flexible coordination, and scalable policy learning [85]. The primary challenges lie in optimizing inter-level coordination, ensuring efficient policy transfer, and balancing local and global decision-making. Addressing these issues is crucial for improving MARL adaptability in large-scale, complex environments.

4) Causal Inference: There are multiple complex variables in MARL, and it is usually difficult to directly discover their internal relationships. Therefore, some recent research has begun to combine MARL with causal inference to discover the causal relationships between agents or variables, and further understand the operating mechanism of agents through intervention and inference, thereby motivating the agent to conduct more targeted learning [86]. Pearl's three-level causal model is considered a powerful tool for constructing causal relationships between variables [87]. Causal influence reward methods promote the collaborative performance of agents by rewarding those agents that have a causal influence on the actions of other agents, where this causal influence is evaluated by counterfactual reasoning [88]. Pina et al proposed the independent causal learning (ICL) algorithm to evaluate the causal effect of each agent's observations on team collaboration performance and solve the credit allocation problem in the independent learning framework [69].

There are also other works that combine causal inference from different perspectives under the framework of MARL. LAIES mathematically define the concept of fully lazy agents and teams by calculating the causal effect of their actions on external states using the do-calculus process, which solved the sparse reward problem in MARL [70]. FD-MARL allows agents to modify communication messages by choosing the counterfactual that bears the most significant influence on others, the continuous communication based on causal analysis enables efficient information transformation in a fully decentralized manner [67]. The deconfounded value decomposition (DVD) method investigates value function decomposition from the perspective of causal inference, which cuts off the backdoor confounding path from the global state to the joint value function [68]. Additionally, deep motor system [71] introduces individual intention inference and causal reasoning to enable fully decentralized coordination, allowing agents to disentangle other agents from their environment and improve cooperative behavior without relying on CTDE. Causal inference has also applied to some practical application scenarios, such as traffic signal control [89] and human-computer interaction decision-making [90].

III. Towards Applications of MARL in CAV Control

CAV control has obtained substantial interest for its ability to reshape modern transportation, enhancing efficiency, safety, and sustainability. CAVs interact with multiple agents, enabling cooperative driving strategies such as platooning, merging, and intersection navigation, which collectively optimize traffic flow and reduce congestion [102]. As CAV technology advances, the integration of MARL has become increasingly important, allowing vehicles to develop adaptive, cooperative policies for dynamic traffic scenarios.

To systematically explore how MARL contributes to CAV control, we categorize cooperative driving tasks based on the extent of required coordination, forming a taxonomy aligned with fundamental principles of CAV control. Coordination in CAV environments typically involves three key factors: **longitudinal control**, **lateral control**, and **timing constraints**. These aspects define the complexity of interaction, shaping how vehicles learn and adapt to shared environments. Therefore, we define three levels of cooperation complexity in MARL-driven CAV control.

- **One-dimensional cooperation** focuses on longitudinal control, where vehicles primarily regulate speed and following distances, as seen in car-following and platooning tasks.
- **Two-dimensional cooperation** extends this by introducing lateral control, requiring inter-agent negotiation for lane changes, merging, and overtaking, which increases the complexity of coordination.

TABLE II
AN OVERVIEW OF THE PRIMARY WORKS FROM 1-D AND 2-D COOPERATION FOR MARL APPLIED IN CAV CONTROL

Scenarios	Typical application	Work	MARL algorithm	Novelty	Metrics
1-D Cooperation	Platooning control	[91], 2021	C-HARL	• Cooperative hierarchical attention RL (C-HARL) • Meta-learning with GNN integration	• Convergence • Success rate of message • Communication delay
		[92], 2022	MADDPG	• A Stackelberg game to model interactions • Decisions sharing with social partners within the platoon due to social effects	• Convergence • Defined utility value • Social effect
		[93], 2023	Distributed PPO	• State fusion strategy with the equilibrium to stabilize traffic oscillations • A reward function design in quadratic form	• Driving comfort • Stability • Travel efficiency
		[94], 2023	Modified MADDPG	• Task-wise value function decomposition • Adaptive agent-specific policies	• Communication latency •Network throughput
		[95], 2024	MARL-PPO	• Decentralized MARL with PPO • Adaptive reward mechanism	• Safety • Traffic efficiency • Merging stability
		[96], 2024	RPE-MAAC	• Relative position encoding for cooperative control to enhance spatial awareness among agents • HDV trajectory generation with IDM-Informer	• Fuel consumption • Stability • Traffic throughput
2-D Cooperation	Cooperative lane changing	[97], 2022	QMIX	• The simplification of system complexity through RL • The fairness in cooperation to allow for independent lane-changing and overtaking	• Collaborative effect • Travel efficiency
		[98], 2022	Bi-level DQN	• An innovative reward function designed with their own benefits and the impact on overall traffic • The driving intentions of the surrounding vehicles encoded into the observation space	• Comfort • Safety • Travel efficiency
		[99], 2022	MA2C	• A innovative local reward and parameter-sharing • Multi-objective reward function	• Driving comfort • Safety • Travel efficiency
		[100], 2024	DQNRR	• Request–respond mechanism for lane-changing coordination • Hierarchical multi-Agent structure	• Traffic throughput • Collision rate
		[101], 2024	PAE-MARL	• Prioritized action extrapolation (PAE) mechanism • A vectorized input representation that condenses vehicle state, scene elements, and risk information	• Collision rate • Average speed
		[29], 2025	FAIRLANE	• Decentralized adaptive lane management • Pre-training distinct policies and assigning them to agents dynamically	• Lane change frequency • Waiting time

- **Three-dimensional cooperation** incorporates time constraints, encompassing tasks such as traffic signal coordination, intersection management, and on-ramp merging, where spatial and temporal synchronization becomes essential.

We categorize MARL advancements by control components and review key simulation platforms for CAV applications, with a focus on cooperative control.

A. One-Dimensional Cooperation

In this subsection, we provide a detailed review of one-dimensional cooperation applications, primarily focusing on longitudinal control. A key example is **platooning control**, where CAVs operate in tightly coordinated formations to improve efficiency, safety, and fuel economy. By minimizing aerodynamic drag and optimizing traffic flow through synchronized control and inter-vehicle coordination, platooning exemplifies how MARL enhances the capabilities of CAVs.

1) Application Scenarios: As shown in Fig. 4, the problem of *platooning control* is commonly addressed through a model-free, multi-agent network approach [58]. In this framework, each agent, symbolizing an AV, has the ability to communicate with both preceding and following vehicles via V2V communication channels. This setup aims to ensure safe, fuel-efficient, and smooth vehicle following operations, while also maximizing the advantages of driving in close formation [103].

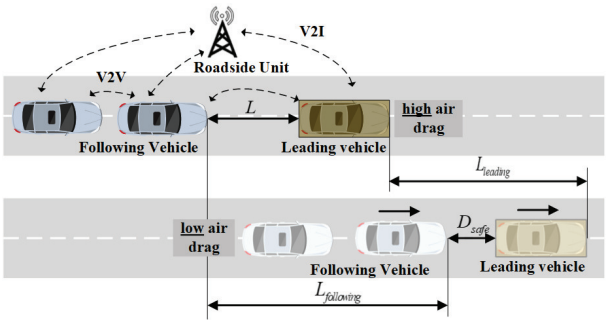

Fig. 4. One-dimensional cooperation scenarios: the case of platooning control.

Given N vehicles in a platoon, the control objective is to maintain a desired spacing L_i and ensure velocity matching. The state vector for each vehicle i at time t is:

$$\dot{\mathbf{x}}_i(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x}_i(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \big(k_p\left((x_{i-1}(t) - x_i(t)) - L_i\right) + k_d\left(v_{i-1}(t) - v_i(t)\right)\big) \quad (1)$$

where $x_i(t)$ is the position of the *ith* vehicle at time t, $v_i(t)$ is the velocity of the *ith* vehicle at time t, L_i is the desired inter-vehicle distance (spacing) between vehicle i and $i+1$, $\mathbf{x}_i(t) = [x_i(t), v_i(t)]^T$ is the state vector, k_p and k_d are the proportional gain for position error and the derivative gain for velocity error respectively. To optimize the platoon control, an objective function J can be defined, incorporating both spacing error and velocity error over a time horizon T:

$$J = \sum_{i=1}^{N} \int_0^T \left(\alpha e_{x,i}^2(t) + \beta e_{v,i}^2(t)\right) dt \tag{2}$$

where α and β are weighting factors that balance the importance of position and velocity errors. Thus, cooperative strategies would be developed to achieve the collective goal, adapting their behaviors in response to the actions of other vehicles within the network.

Mixed platooning control refers to scenarios where multiple vehicle platoons coexist within a network, requiring sophisticated coordination and communication both within (intra-platoon) and between (inter-platoon) these formations [104]. To address these challenges, Parvini et al. introduced two advanced MARL-based algorithms specifically designed for mixed platooning control. One of these, Modified MADDPG, along with its task decomposition variant, improves resource allocation in platoon-based cellular V2X (C-V2X) networks. By employing task-wise value function decomposition and adaptive agent-specific policies, this method effectively reduces communication latency and optimizes network throughput, leading to more efficient vehicular coordination [94]. In these frameworks, platoon leaders function as independent agents that interact with the environment to determine optimal policies for platoon management. These algorithms leverage multiple critics to estimate both global and local rewards, fostering cooperative behavior among agents. Additionally, the task decomposition variant further refines individual reward structures, breaking them down into task-specific sub-reward functions to enhance coordination and policy learning, as demonstrated in [94].

Even though CAVs have made significant advancements, there will be an extended period during which CAVs will coexist with human-driven vehicles (HDVs), a situation commonly referred to as *mixed-traffic* scenarios [45]. Different penetration rates can be assessed to understand the impact of mixed traffic. In the study presented in [105], the impact of the penetration rate of CAVs on the energy efficiency of the traffic network is examined, with a specific emphasis on a cooperative eco-driving system that utilizes longitudinal control. Lu et al. enhance the evaluation by integrating several key elements, including altruism control, a quantitative car-following strategy, a refined platoon reward function, and a collision avoidance method, as outlined in their work [106]. In [93], the authors introduce the concept of characterizing consecutive HDVs as a collective entity, referred to as AHDV, to minimize stochastic variability and leverage macroscopic characteristics for the control of following CAVs. They employ a control strategy built on distributed proximal policy optimization (DPPO) [107] to anticipate disturbances and downstream traffic conditions in mixed traffic scenarios. Additionally, Relative Position Encoding Multi-Actor Attention Critic (RPE-MAAC) [96] introduces relative position encoding to enhance spatial awareness and cooperative control in mixed CAV-HDV platoons. Integrated with the Intelligent Driver Model-Informer (IDM-Informer), RPE-MAAC generates realistic HDV trajectories, mitigating error propagation while improving fuel consumption, stability, and overall traffic throughput. In vehicular environments characterized by high mobility, the use of traditional centralized optimization methods that rely on global channel information can be impractical. In [108], the authors model each CAV as an individual agent, which makes decisions based on local information and communication with neighboring vehicles, without relying on a centralized controller.

2) Optimization Goals: One critical goal of platooning control is to achieve *string stability*, which refers to the ability of a line of CAVs to maintain a stable and orderly formation (e.g., stable distance and speed) as they travel in close proximity to each other [109]. Notably, MARL has been proposed to address and enhance the achievement of string stability in the platooning control scenario, contributing to the development of more efficient and coordinated platooning control strategies. For instance, in [110], a MARL approach based on a robust communication protocol with Long Short-Term Memory (LSTM) [111] is introduced, resulting in stable platooning control. In addition, Li et al. introduce the Communication Proximal Policy Optimization (CommPPO) algorithm to enhance stability, which adapts to varying agent numbers and supports different dynamics of platooning control [112]. CommPPO incorporates a predecessor-leader-follower communication protocol to facilitate the transmission of both global and local state information among agents. Notably, CommPPO introduces a novel reward communication channel, effectively mitigating issues related to spurious rewards and mitigating the problem of "lazy agents", which are commonly encountered in other MARL approaches. Similarly, Multi-Agent Reinforcement Learning with Proximal Policy Optimization (MARL-PPO) [95] enhances highway platooning stability through decentralized learning and an adaptive reward mechanism, dynamically adjusting inter-vehicle distances based on velocity and lane positioning to improve safety, traffic efficiency, and merging success rates in dynamic traffic conditions. In [58], the platooning control problem is formulated as a spatiotemporal MDP. They enhance system stability by introducing a spatial discount factor for local agents.

Efficient communication among CAVs in platooning control also poses a fundamental task in achieving seamless coordination and maintaining the desired following properties, such as precise inter-vehicle spacing and synchronized maneuvers. For instance, Liu et al. introduce a Multi-Agent Hierarchical Attention RL (MAHARL) framework, which leverages hierarchical Graph Attention Networks to model inter-agent influence and enhance long-term decision-making without immediate rewards [91]. Additionally, in scenarios where complete information is lacking, Li et al. [92] develop a game-theoretic framework to capture the strategic interactions between the leading vehicle and the following vehicles. To

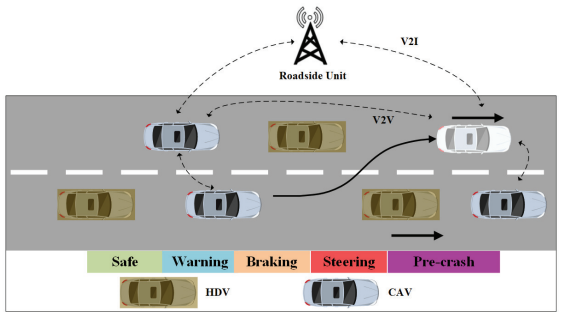

Fig. 5. Two-dimensional cooperation scenario.

overcome the absence of full information, the equilibrium solution in platooning control is identified through backward induction, coupled with information collection from the following vehicles.

B. Two-Dimensional Cooperation

Compared to one-dimensional cooperation, two-dimensional cooperation addresses problems related to both longitudinal and lateral control, with its most typical application being *cooperative lane changing*.

1) Application Scenarios: As shown in Fig. 5, CAVs communicate and coordinate through V2V and V2I communication channels, enabling them to make decisions and acquire effective lane-changing strategies within shared driving environments [113]. Generally, the lane-changing mathematical problem [114] can be expressed as follows:

$$\begin{aligned}\min_{\mathbf{u}_i(t)} J = \int_{t_0}^{t_f} \Bigg[& \alpha_1 \Big((x_{i-1}(t) - x_i(t) - L_i - \Delta x_{\text{safe}})^2 \\ & + \left(x_i(t) - x_{i+1}(t) - L_{i+1} - \Delta x_{\text{safe}}\right)^2 \Big) + \alpha_2 \left(t_f - t_0\right) \\ & + \alpha_3 \Big(u_{x,i}(t)^2 + u_{y,i}(t)^2 + \left(\frac{d}{dt} u_{x,i}(t)\right)^2 \\ & + \left(\frac{d}{dt} u_{y,i}(t)\right)^2 \Big) \Bigg] dt \end{aligned} \tag{3}$$

Subject to the vehicle dynamics:

$$\dot{\mathbf{x}}_i(t) = \mathbf{A}\mathbf{x}_i(t) + \mathbf{B}\mathbf{u}_i(t) \tag{4}$$

where α_1, α_2, and α_3 are the corresponding weights for safety, efficiency, and comfort, respectively. Other objectives may also be considered; $\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\mathbf{x}_i(t) = \left[x_i(t), y_i(t), v_{x,i}(t), v_{y,i}(t)\right]^T$ is the state vector for vehicle i; $x_i(t)$ and $y_i(t)$ are the longitudinal and lateral position of the *ith* vehicle at time t; $v_{x,i}(t)$ and $v_{y,i}(t)$ are longitudinal and lateral velocity of the *ith* vehicle at time t. $\mathbf{u}_i(t) = \left[u_{x,i}(t), u_{y,i}(t)\right]^T$ is the control input vector for vehicle i, $u_{x,i}(t)$ and $u_{y,i}(t)$ are the control input for longitudinal and lateral acceleration respectively; L_i is the length of the *ith* vehicle; Δx_{safe} is the safety distance to avoid collision; w_{lane} is the width of the lane.

And the constraints are expressed as:

$$\begin{aligned} & x_{i-1}(t) - x_i(t) \geq L_i + \Delta x_{\text{safe}} \\ & x_i(t) - x_{i+1}(t) \geq L_{i+1} + \Delta x_{\text{safe}} \\ & y_i(t) \in \{\text{current lane, target lane}\} \\ & 0 \leq y_i(t) \leq w_{\text{lane}} \end{aligned} \tag{5}$$

Hou et al. devise a decentralized *cooperative lane-changing* controller employing a multi-agent Proximal Policy Optimization (MAPPO) [115] approach, which empowers each vehicle to independently learn and assess its policy and action rewards based on local information, while still having access to global state data. The trained policies can be effectively transferred and applied across various traffic conditions, achieving a safe, efficient, and comfortable lane-change maneuver [116]. Dubey et al. [29] introduce FAIRLANE, a MARL-based framework designed to manage priority lanes in mixed-autonomy traffic. By utilizing multi-policy bootstrapping and decentralized decision-making, the framework dynamically optimizes lane allocation to enhance traffic efficiency, safety, and fairness. This method ensures equitable access to priority lanes while adapting to real-time traffic conditions. Zhang et al. [101] propose a Multi-Agent Reinforcement Learning (MARL) framework that incorporates a Prioritized Action Extrapolation (PAE) mechanism. This approach dynamically updates lane-changing decision priorities, enabling smoother coordination, improved safety, and faster policy convergence in high-density traffic scenarios.

In [97], the authors put forth a *multi-vehicle cooperative driving* approach, leveraging the QMIX algorithm. Their method applies RL to dynamically adapt to changing conditions on highways, aiming to find an optimal balance between autonomous decision-making and cooperative interaction among the vehicles. This algorithm allows each vehicle in the pair to independently perform lane changes and overtaking maneuvers, even in dense traffic scenarios, while maintaining a predetermined formation between them. An enhanced QMIX algorithm [117] is further proposed to enhance the flexibility and effectiveness of the collaborative lane-changing system. They implement a stable estimation method to mitigate the problem of overestimated joint Q-values among agents. This approach strikes a fine balance between maintaining formation and allowing for smooth overtaking, thus facilitating intelligent adaptation to a variety of scenarios, including heavy traffic, light traffic, and emergencies. In an effort to enhance mobility within complex traffic environments, in [118], instead of relying on relative distance and semantic maps, the authors suggest the utilization of traffic states that encapsulate the spatio-temporal interactions between neighboring vehicles. Within the MADRL framework, three prediction models, namely the transformer-based (TS-Transformer), generative adversarial network-based (TS-GAN), and conditional variational autoencoder-based (TS-CVAE) models, are developed and compared for traffic state prediction.

2) Optimization Goals: Safety is of paramount importance in the context of CAV scenarios. In [119], the authors have introduced a safety-enhancing actor-critic algorithm that incorporates two innovative techniques. The first technique is

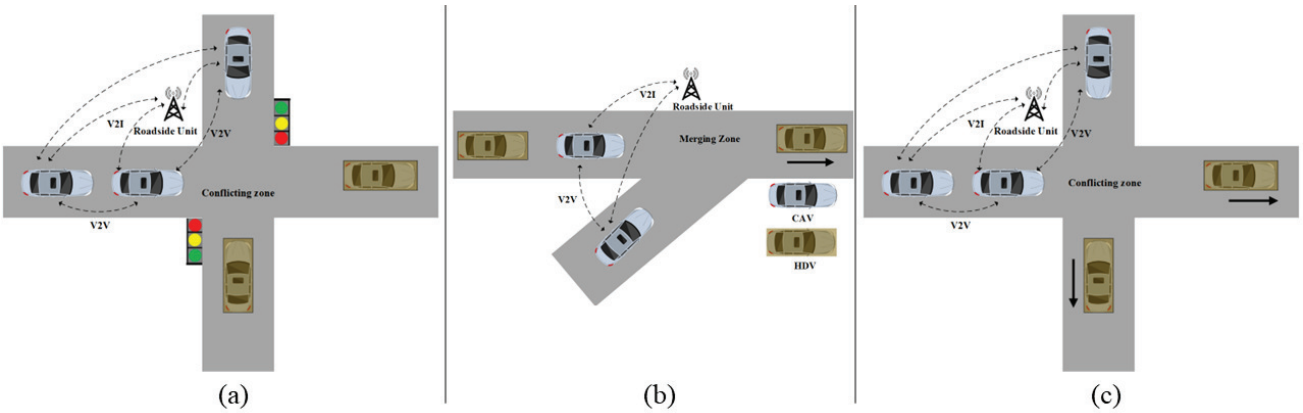

Fig. 6. Three-dimensional cooperation scenarios: (a) Traffic Signal Coordination; (b) Merging on-ramps;(c) Unsignalized intersections.

the 'truncated Q-function', which effectively leverages shared information from neighboring CAVs, ensuring scalability for large-scale CAV systems. The second technique, "safe action mapping", provides safety guarantees throughout both the training and execution phases by utilizing control barrier functions. Additionally, a bi-level strategy is proposed in [98] to further enhance safety and efficiency. At the upper level, the MADQN model is utilized for making decisions about lane changes. This approach acknowledges the cooperative aspect of driving by factoring in the intentions of surrounding vehicles, enabling implicit negotiation for right-of-way. At the lower level, a right-of-way assignment model is utilized to ensure safety. A novel reward function is further proposed to encourage coordination and account for traffic impact. Moreover, Li et al. have introduced MetaDrive, a platform that focuses on safe driving by addressing generalizability, safety awareness, and multi-agent decision-making. It offers diverse driving scenarios for benchmarking single and multi-agent RL tasks [120].

Graph neural networks (GNNs) [121] have gained significant attention in MARL settings, particularly for *traffic flow optimization*. Chen et al. introduce a novel algorithm based on MADQN, which combines a GCN with a deep Q-network [122]. This approach facilitates effective information fusion and decision-making within the MARL framework, thereby enhancing both safety and traffic flow in various traffic scenarios. Moreover, in [123], the authors leverage GNN in conjunction with MADDPG in their work for multi-agent training, addressing the complexity of real-world inputs and aiding in congestion mitigation efforts. Guo et al. introduce deep Q-Network with Request-Respond Mechanism (DQNRR), which enhances lane-changing coordination through a request-respond mechanism. This method hierarchically structures lane-change decisions, allowing vehicles to negotiate merging and overtaking maneuvers more efficiently, thus reducing computational complexity while improving lane-changing efficiency in congested urban traffic [100].

Thoughtful reward function design is paramount for the *adaptivity and efficiency* of lane-changing strategies. In [99], the authors introduce the multi-agent advantage actor-critic (MA2C) framework, which integrates a novel local reward design and parameter-sharing scheme for cooperative lane changing in mixed traffic scenarios. They also present a multi-objective reward function that considers factors such as fuel efficiency, driving comfort, and safety to facilitate successful and efficient lane changing. In [124], a MARL approach is explored in which agents collaborate to reach a zero-sum game state. In contrast to cooperative driving, this approach, known as harmonious driving, places emphasis on achieving a balance between overall and individual efficiency while utilizing limited sensing data from individual vehicles. They design a reward function that promotes harmony by considering both individual and overall efficiency, as opposed to competitive strategies that solely prioritize individual interests.

C. Three-Dimensional Cooperation

Three-dimensional cooperation involves the intricate coordination of longitudinal and lateral movements along with timing, across a variety of traffic scenarios such as traffic signal coordination, on-ramp merging, and navigating through unsignalized intersections, as depicted in Fig. 6. To ensure safe, efficient, and effective cooperation, CAVs need to rapidly learn longitudinal and lateral maneuvers within tight time constraints. The three-dimensional cooperation problem can also be described by Eq. 3–Eq. 5, with the added time constraints $t_{i,\text{ start}} \leq t \leq t_{i,\text{ end}}$.

1) Traffic Signal Control (TSC): TSC plays a critical role in shaping vehicle interactions by introducing time-based constraints at intersections (see Fig. 6 (a)). Traditional optimization-based TSC methods focus on traffic management, adjusting signal timings to regulate flow, but they treat vehicles as passive entities responding to fixed control policies [45]. In contrast, when considering MARL in CAV control, TSC is not the primary research target but rather an external constraint that impacts how CAVs cooperate to navigate intersections efficiently. CAVs, unlike human-driven vehicles, can actively communicate and coordinate their movements under these constraints, optimizing their joint decision-making to minimize congestion, delays, and energy consumption. MARL provides a natural framework for CAVs to learn cooperative policies in response to signal timings, ensuring efficient merging, gap acceptance, and right-of-way negotiation. The challenge

lies not in optimizing TSC itself but in understanding how CAVs interact under these time-dependent traffic conditions to improve overall system efficiency. TSC can be viewed as a multi-objective optimization problem and Markov/Stochastic Games with cooperative settings, which require agents to coordinate with each other. Therefore, the learning-based method can show better performance without prior knowledge about the given environment. In multi-agent-based algorithms, all traffic light controllers within a traffic grid need to coordinate to address traffic congestion.

Value-based MARL algorithms have been extensively investigated for TSC challenges. For instance, to alleviate the curse of dimensionality and environmental nonstationarity problems, a decentralized coordination MADQN approach is proposed in [125] to explicitly identify and dynamically adapt agent coordination needs during the learning process. Similarly, in [126], the complex TSC problem is decomposed into simpler subproblems and tackled using multiple regional agents and a centralized global agent. Each regional agent learns its RL policy for smaller regions with reduced action spaces, while the centralized global agent combines the RL contributions from various regional agents to form a final Q-function for the entire large-scale traffic grid. Notably, QMIX, an extension of VDN, was investigated in [127] for TSC, achieving promising results. Additionally, Liu et al. introduce a multi-agent dueling-double-deep Q network (MAD3QN) for TSC [128]. This framework features an innovative γ-reward design, which combines the traditional γ-reward with a novel γ-attention-reward. By employing a spatial differentiation method for agent coordination, their approach enables decentralized control and decoupling of road networks, enhancing scalability and convergence. In [129], a cooperative group-based multi-agent Q-learning for ATSC (CGB-MATSC) is proposed. This method enhances the learning process by incorporating a k-nearest-neighbor approach for state representation, a pheromone-based strategy for creating regional green-wave traffic flows, and a spatially discounted reward system.

On the other hand, *policy-based* MARL algorithms have also been widely studied for TSC. In [36], a fully scalable and decentralized Multi-Agent Actor-Critic (MA2C) algorithm is introduced, which incorporates an advanced discount factor to mitigate the effects of remote agents. This approach demonstrates significant advantages over random and independent controllers. Wu et al. introduced the multi-agent recurrent deep deterministic policy gradient (MARDDPG) algorithm, specifically designed for TSC in vehicular networks [130]. They utilize a strategy of CTDE, enhancing the efficiency of the training process through parameter sharing among the actor networks. Furthermore, the integration of LSTM networks enables the algorithm to utilize historical data for more effective control decisions. In [131], a MA2C approach with a feudal hierarchy concept for TSC is presented. This approach segments the traffic network into several regions, each monitored by a manager agent, with traffic signal agents acting as workers. The managers are responsible for high-level coordination and setting objectives for the workers, who then adjust traffic signals to meet these objectives. This hierarchical structure fosters global coordination while ensuring the system scalability.

Efficient *communication* is crucial in applying MARL to TSC as it enables coordinated decision-making, optimal resource allocation, adaptation to dynamic environments, scalability, conflict resolution, and efficient learning, all of which contribute to effective traffic management [132]. Liu et al. present a novel algorithm that enhances communication efficiency through a new message exchange method and improves congestion measurement by introducing a more comprehensive reward calculation method. Then a clear and simple representation of traffic conditions is provided, and the synchronization between agents at different intersections with varying cycle lengths needs to be addressed [133]. To incorporate spatial and temporal dependencies, Wang et al. introduce STMARL [134], which models traffic lights using a traffic light adjacency graph and recurrent neural networks. This approach optimizes coordination across intersections by merging historical and real-time traffic data. Recognizing the importance of shared information, Jiang et al. propose UniComm [135], which predicts the impact of local observations on neighboring intersections, improving communication efficiency.

Graph-based MARL provides advantages such as scalability, efficient message passing, global coordination, and adaptability, making it particularly suitable for multi-agent traffic signal control (TSC) problems [136]. CoLight [137] introduces a graph-attentional MARL model that enables traffic signals to communicate using temporal and spatial data from neighboring intersections. By achieving index-free modeling of intersections, it enhances adaptability and efficiency. Similarly, Yang et al. propose Inductive Heterogeneous Graph Multi-agent Actor-Critic (IHG-MA) [138], which leverages an inductive heterogeneous graph neural network to encode traffic network structures and a multi-agent actor-critic framework for policy learning, allowing real-time adaptation to previously unseen intersections. Further enhancing MARL-based intersection control, Antonio et al. [139] integrate multi-agent TD3 with LSTM to manage varying observation sizes due to fluctuating vehicle numbers. Their method employs curriculum learning through self-play to optimize collaborative CAV control at intersections, improving traffic efficiency and safety. In contrast, Ren et al. highlight potential security vulnerabilities in MARL-based traffic signal control by proposing a stealthy opaque-model attack strategy. Their method selectively perturbs critical states to increase congestion and delays while remaining undetectable, underscoring the need for robust security mechanisms in intelligent transportation systems [140].

2) On-Ramps Merging: The on-ramp merging task requires CAVs to integrate into the main traffic flow while maintaining safety and efficiency (see Fig. 6 (b)). This maneuver involves both spatial and temporal coordination, as vehicles must dynamically adjust their speeds and trajectories to avoid conflicts. CAVs in the through lane proactively modify their velocities—either accelerating or decelerating—to create sufficient gaps for merging vehicles. Meanwhile, CAVs on the ramp must regulate their speeds and make timely lane-change decisions to merge smoothly without causing congestion. The success of this process depends on precise multi-agent

TABLE III
AN OVERVIEW OF THE PRIMARY WORKS FROM FOUR PERSPECTIVES IN DIFFERENT SETTINGS

Scenarios	Typical application	Work	MARL algorithm	Novelty	Metrics
3-D Cooperation	Traffic signal control	[128], 2021	MADQN	• The γ-reward method to extend the Markov chain to the space-time domain • The spatial differentiation method to amend the current reward by recursion	• Asymmetry road network evaluation • Attention score • Training time
		[129], 2021	CG-Based MA-Q Learning (CGB-MAQL)	• Agents organized into cooperative groups with each agent responsible for its region • A k-nearest-neighbor-based joint state representation among neighboring agents • A heuristic training mechanism by terminating poor behavior strategies	• Waiting time • Road pheromone
		[133], 2023	MA-Q Learning	• A new message sending and processing • A data structure to record the latest and most valuable message • A new reward calculation with both queue length and total waiting time	• Waiting time • Reward
		[140], 2024	SBA-DT	• An attack method that perturbs critical states in a black-box setting • A new agnostic dynamic threshold-based defense mechanism • Multi-objective disruption strategy	• Traffic efficiency • Attack effectiveness
	On-ramps merging	[144], 2022	MADDPG	• Transferring the policies through a few learning CAVs in the training process • Reward function design at high speeds	• Traffic flow smoothness
		[143], 2022	Distributed MADDPG	• The interference of human-driven Vehicles (HDV) • Reward function design including rear-end safety, lateral safety, and energy consumption	• Fuel consumption • Travel time
		[30], 2023	MA2C	• An action masking scheme • Parameter sharing and local rewards to encourage inter-agent cooperation • A novel priority-based safety supervisor introduced to reduce collision rates	• Traffic throughput • Collision rate
		[20], 2024	GCAV-CPO	• A innovative local reward and parameter-sharing • Multi-objective reward function	• Driving comfort • Safety • Travel efficiency
	Unsignalized intersections	[19], 2022	MADDQN	• Route-agents introduced to coordinate the actions of multiple vehicles • The inclusion of a collision term in the reward function enables cooperation • The reuse of knowledge from agents' policies to handle unknown traffic scenarios	• Safety • Travel time • Fuel consumption
		[147], 2023	QMIX	• Integration expert demonstrations into MARL training • Adaptive interaction-based learning	• Travel time • Success rate • Average travel time
		[19], 2024	RA-CoMARL	• Route-agent framework for efficient learning • A novel collision term incorporated into the state space	• Collision rate • Average velocity • Fuel consumption
		[18], 2024	MA-GA-DDPG	• Interactions and negotiations model establishment between multiple agents • An experience replay mechanism to tackle the non-stationarity	• Timeout • Accident • Success rate

coordination, where vehicles anticipate and adapt to each other's actions, ensuring safe and efficient merging under real-time traffic constraints [141].

In [142], a simplified mathematical formulation for an on-ramp merging scenario is presented to model the fundamental interactions and solved with a MADQN algorithm that accounts for the interaction between an on-ramp merging vehicle and a traffic vehicle in the target lane. The results indicate that a multi-agent approach can result in reduced collision rates compared to a single-agent approach, but this improvement is contingent on the optimality of the traffic vehicle's behavior in the target lane. Zhou et al. introduce a cooperative merging control strategy for CAVs using a distributed MADDPG approach, which accounts for safe merging distances, acceleration limits, and factors like rear-end safety, lateral safety, and energy consumption [143]. To address the challenge of a dynamic environment resulting from the decentralized learning of CAVs, a decentralized framework employing MADDPG is presented to coordinate CAVs during highway merging [144]. This framework enables

policies learned from a small group of trained CAVs to be transferred and applied to any number of CAVs, with a reward function that promotes high-speed travel, resulting in smoother traffic flow while maintaining safety in terms of rear-end and lateral collisions.

Some studies also examine the influence of *human drivers* in on-ramp merging scenarios emphasizing adaptive decision-making and cooperative control between connected and automated vehicles (CAVs) and human-driven vehicles (HDVs). Hu et al. propose Interaction-aware Decision-making with Adaptive Strategies (IDAS), which leverages a modified Multi-Agent Advantage Actor-Critic (MA2C) method and curriculum learning to train AVs to adapt to varying driver behaviors by assessing their level of cooperativeness [145]. Similarly, Chen et al. introduce a multi-agent cooperative merging framework using MA2C, enabling CAVs on both the merge and through lanes to jointly develop policies that optimize traffic flow and adapt to HDV dynamics [30]. Their framework ensures scalability in dynamic traffic scenarios through parameter sharing, local rewards, and action masking, promoting inter-agent cooperation while incorporating a priority-based safety supervisor to reduce collision risks and accelerate learning.

Beyond RL approaches, game-theoretic strategies have also been explored to enhance *safety*. Chandra et al. propose GAMEPLAN, a game-theoretic multi-agent planning approach that integrates both CAVs and human drivers in merging scenarios [146]. This method utilizes game theory and an auction-based mechanism to compute optimal driving actions while inferring driving styles from sensor data. By assigning priority based on driver aggressiveness or patience, GAMEPLAN optimally balances safety and efficiency, preventing both collisions and deadlocks.

To tackle the complexities of multilane on-ramp merging, Liu et al. introduce Graph Convolutional Proximal Policy Optimization (GCAV-CPO), a *graph-based MARL* framework that integrates vehicle graph structures with an eco-friendly Markov decision process to enhance coordination, safety, and energy efficiency in mixed-autonomy traffic [20]. Their approach generalizes well across varying traffic densities and CAV penetration rates, demonstrating superior performance over existing baselines.

3) Unsignalized Intersections: Intersections serve as pivotal nodes in urban road networks but can also become major bottlenecks if not managed efficiently. However, managing vehicular interactions in the absence of traffic signals, as illustrated in Fig. 6 (b), introduces significant challenges due to the need for decentralized decision-making, right-of-way negotiation, and collision avoidance [19]. Unlike TSC, where vehicles follow fixed timing rules, unsignalized intersections require real-time adaptive cooperation among multiple CAVs. Many conventional approaches, such as rule-based [148], planning-based [149], and single-agent RL methods [147], often treat intersection management as a single-agent problem, which can not capture the complex interactions among CAVs. Recently, MARL has emerged as a promising and effective tool for managing traffic at unsignalized intersections.

Spatharis et al. propose a multi-agent double DQN framework for unsignalized intersections, integrating an efficient reward function to balance safety and traffic efficiency while enabling transfer learning for adaptability [19]. Capasso et al. develop a MAD-A3C model using model-free DRL to predict vehicle actions, allowing CAVs to autonomously learn intersection navigation while ensuring safety [150]. Yan et al. introduce a DQN-based policy decomposition strategy for decentralized CAV coordination, achieving near-optimal throughput without reward shaping and demonstrating strong generalizability across traffic conditions [151].

Effective *coordination* among CAVs is essential to ensure safe and efficient maneuvers at unsignalized intersections. In [152], a decentralized MADQN algorithm is adopted and learned to make decisions for CAVs. To enable effective coordination among agents, the intent trajectories of other neighboring agents are incorporated into each agent's state space. Furthermore, a decentralized and conflict-free coordination scheme designed for CAVs at unsignalized intersections is proposed in [153], with the objective of enhancing intersection management precision. They frame the challenge of safely guiding multiple vehicles through unsignalized intersections as a partially observable stochastic game (POSG). To facilitate collaborative decision-making in a distributed fashion, they introduce a cooperative multi-agent proximal optimization algorithm (CMAPPO). In [154], a multi-layer coordination strategy is proposed for the management of unsignalized intersections. This architecture comprises two layers: a low-level layer that employs a dynamics-based algorithm to control individual CAVs, and a high-level layer that integrates a TD3 algorithm, trained in a centralized manner but executed in a decentralized fashion by multiple agents for decision-making. Their findings reveal a successful training process, with the MATD3 algorithm achieving a remarkable 100% success rate in preventing intersection collisions. Addressing adjacent intersection coordination, Xu et al. propose a multi-agent deep reinforcement learning scheduling (MA-DRLS) algorithm, allowing each intersection agent to independently optimize traffic scheduling via information exchange, using DQN networks for stable training [155]. Additionally, Tallapragada et al. introduce a multi-agent joint-action DDPG (MAJA-DDPG), a framework integrating RL with sequential optimization. It first learns a shared policy for determining vehicle crossing orders based on traffic states, followed by trajectory optimization, ensuring scalable and distributed coordination [156].

To enhance coordination in bandwidth-constrained multi-agent settings, Li et al. introduce the Efficient Communication Method (ECM)-MA2C algorithm [157]. By integrating a variational auto-encoder with a multi-head attention mechanism, ECM-MA2C selectively extracts and retains critical information from neighboring agents, thereby optimizing communication efficiency. Experimental results demonstrate that this approach significantly outperforms baseline methods, ensuring effective coordination with minimal communication overhead. While ECM-MA2C focuses on efficient information sharing, Guo et al. [158] tackle decision-making challenges at unsignalized intersections using a value decomposition-based

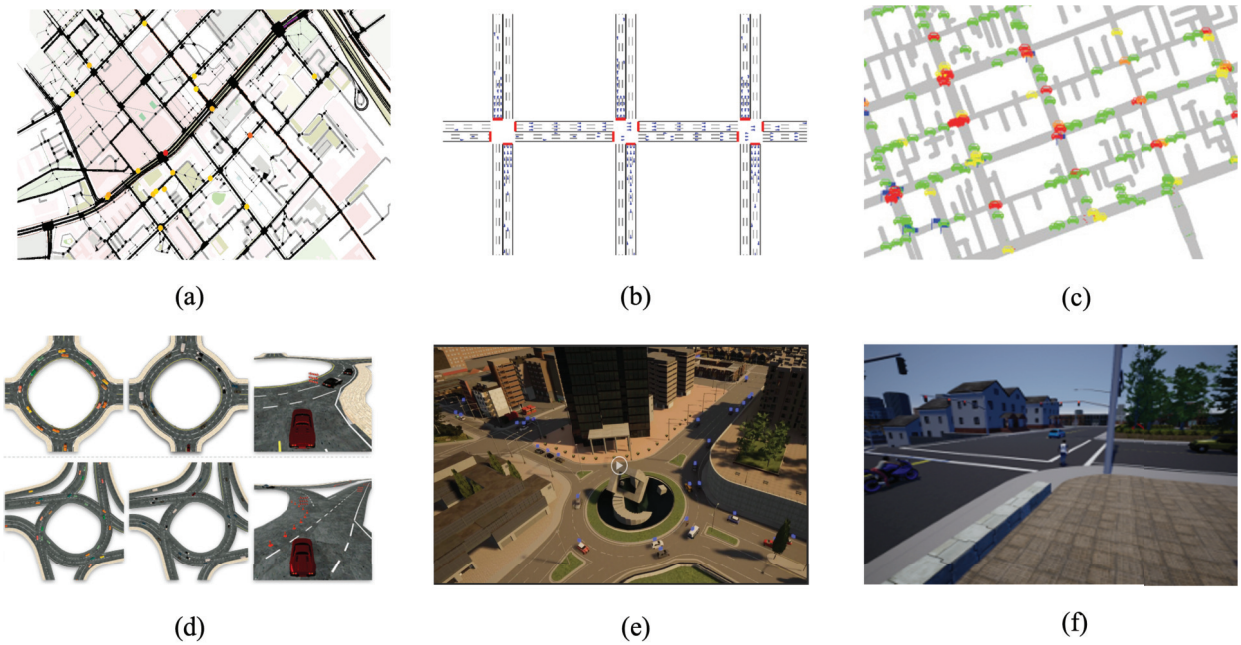

Fig. 7. Representative scenarios in different simulators: (a) SUMO [159]; (b) CityFlow [160]; (c) SMARTS [161]; (d) MetaDrive [120]; (e) CARLA [162]; (f) MACAD [163].

MARL framework, QMIX. Their enhanced QMIX implementation introduces network-level optimizations, TD updates with discount factor γ, and reward clipping techniques, all of which contribute to improved learning stability and more reliable cooperative control. To further improve learning efficiency in complex intersection scenarios, Spatharis et al. [19] proposed a Route-Agent Cooperative Multi-Agent Reinforcement Learning (RA-CoMARL) framework for autonomous driving at unsignalized intersections, which introduces route agents as a scalable decision-making structure and a collision prediction mechanism to enhance traffic coordination, reduce congestion, and improve safety. Building upon these advancements, Liu et al. [18] introduce the Multi-Agent Game-prior Attention DDPG (MA-GA-DDPG) algorithm. This framework integrates an attention mechanism to capture inter-vehicle dependencies while incorporating a hierarchical game-theoretic structure to enhance safety, efficiency, and learning stability. By leveraging these mechanisms, MA-GA-DDPG ensures robust and scalable cooperative decision-making for CAVs at unsignalized intersections, paving the way for more adaptive and resilient intersection management strategies.

D. Simulation Platforms

The simulation platforms, which act as the training/learning environments, play an important role in the performance and adaptability of the autonomous agents. The learning platforms provide different challenges for CAV control by considering different traffic scenarios and different characters of agents (Fig. 7). The traffic ecosystem/simulation platform incorporates realistic models, physics-based simulations, and efficient numerical solvers, providing a controllable environment to accelerate the development of CAVs [164]. Comparisons of different simulation platforms for CAVs are shown in Table IV.

SUMO (Simulation of Urban MObility) is an open-source, large-scale traffic simulator widely used for studying platooning (1D), lane coordination (2D), and traffic flow optimization (3D) [159]. It supports customizable road networks, real-time traffic data integration, and reinforcement learning experiments, making it a versatile tool for evaluating CAV interactions.

CityFlow is a high-performance traffic simulator designed for large-scale, city-wide coordination [160]. With its efficient data structures and multi-agent RL compatibility, it excels in optimizing TSC (3D) and large-scale intersection coordination while maintaining computational efficiency.

SMARTS (Scalable Multi-Agent RL Training School) is a multi-agent RL platform tailored for cooperative CAV learning [161]. It provides realistic traffic behaviors, customizable scenarios, and benchmark tasks, making it ideal for studying CAV interactions at intersections (3D) and lane coordination (2D) under mixed-traffic conditions.

MetaDrive offers infinite procedurally generated driving environments, balancing realism with efficiency [120]. It is particularly well-suited for studying generalizable RL in highway merging (2D) and intersection scenarios (3D) while maintaining lightweight computational requirements.

CARLA, is a high-fidelity 3D driving simulator with realistic physics and sensor modeling, enabling MARL research in complex urban settings [162]. It supports CAV coordination under dynamic environments, making it ideal for advanced intersection management (3D) and adaptive lane merging (2D).

MACAD (Multi-Agent Connected and Autonomous Driving) provides a multi-agent RL testbed for CAVs based on partially observable Markov games [163]. It supports cooperative intersection navigation (3D) and deep RL-based vehicle interaction studies, making it a valuable tool for high-level CAV coordination research.

TABLE IV
COMPARISONS BETWEEN SOME TYPICAL CAV CONTROL SIMULATION PLATFORMS

Simulators	Control Dimensions	Vehicle Dynamics	3D Scenarios	Visual Rendering	Pedestrians	Light Weighted	Customized Maps	Data Importing
Highway-env [167]	1-D	-	-	-	-	✓	✓	-
Nocturne [166]	2-D	-	-	-	✓	✓	-	✓
SUMO [159]	1/2/3-D	-	-	-	✓	✓	✓	✓
CityFlow [160]	2/3-D	-	-	-	-	✓	✓	✓
SMARTS [161]	2/3-D	✓	-	-	-	✓	✓	-
MetaDrive [121]	1/2/3-D	✓	✓	-	-	✓	✓	✓
CARLA [162]	2/3-D	✓	✓	✓	✓	-	✓	✓
MACAD [163]	2/3-D	✓	✓	✓	✓	-	✓	✓

More simulators for CAVs have been designed to consider more detailed characters and versatile scenarios, which provides more tools to mitigate the gaps between the simulation and reality.

A **PCMA** (pedestrian crash avoidance mitigation) system is a MARL-based system designed to study CAV-pedestrian interactions in unmarked crosswalks [165]. By modeling pedestrian behaviors with both predefined and DRL-based adaptive strategies, it evaluates collision rates, traffic flow efficiency, and observation uncertainty, enabling safer and more adaptive CAV decision-making.

Nocturne is a 2D driving simulator specifically designed for studying partial observability in multi-agent driving [166]. It incorporates real-world data-driven scenarios and efficiently simulates the limited visibility conditions that CAVs encounter, such as occlusions at intersections or sensor limitations.

Generally speaking, the simulation-based evaluations are simpler to implement, reproduce, and scale compared with real experiments for learning. However, simulations may not capture all the challenges associated with an actual deployment. For example, factors such as network delay, vehicle model discrepancies, computation time, and the necessity of implementing clock synchronization and fail-safe routines pose challenges in real-world implementations.

IV. DISCUSSION & FUTURE WORK

CAVs leveraging MARL algorithms present a promising frontier in the evolution of transportation. As urban centers grapple with traffic congestion and road safety, the nuanced and adaptive behaviors offered by MARL algorithms can enable CAVs to respond more fluidly to the actions of other road participants. The dynamism of MARL allows these vehicles to cooperatively optimize traffic flows, potentially alleviating congestion and reducing transit times. Nevertheless, the journey to this future is not without its challenges. Interactions with unpredictable human drivers, managing the delicate balance between exploration and safety in learning, and navigating a complex regulatory landscape are just a few of the hurdles. Yet, with ongoing collaborations among tech giants, startups, and academic circles, there is a collective push to overcome these barriers, signaling an optimistic trajectory for the marriage of MARL and CAVs.

A. Safety Challenges

1) Current Progress: Safety is a fundamental concern in the deployment of MARL for CAVs, as it directly affects public trust, regulatory approval, and real-world applicability. Current research has explored various safety-enhancing strategies, including safety-constrained policy training, cooperative decision-making frameworks, and hierarchical MARL methods [168]. Some approaches enforce hard constraints or incorporate barrier functions into MARL training to restrict unsafe actions, while others modify reward functions to penalize high-risk behaviors [169]. Additionally, graph-based learning has been leveraged to enhance inter-vehicle coordination, reducing the risk of conflicts in dense traffic scenarios. Safe MARL frameworks have also been introduced for specific driving situations, such as lane merging, intersection coordination, and emergency braking scenarios, where explicit safety constraints can be embedded into agent interactions. However, despite these advancements, the majority of existing methods rely on empirical safety validation rather than theoretical guarantees, making their applicability to real-world traffic conditions uncertain. Furthermore, the challenge of generalization to diverse and unseen scenarios persists, as most safety mechanisms are designed and evaluated in constrained simulation settings rather than in open-ended, adversarial, or extreme conditions [170].

2) Research Gaps: Despite the progress in safety-aware MARL for CAVs, several fundamental challenges remain unresolved. First, formal safety guarantees are largely absent, as most existing safety frameworks rely on heuristic-based reward shaping, risk-sensitive training, or pre-defined constraints that do not provide mathematically rigorous assurances of policy safety. To achieve provable safety, future research must explore *control-theoretic safety verification*, *distributionally robust optimization*, and formal methods that can certify MARL policies under *worst-case scenarios*. Second, uncertainty in real-world driving environments—including stochastic traffic behaviors, sensor noise, adversarial interactions, and rare edge cases—poses a major obstacle to safe MARL policy

learning [171]. While some methods incorporate robust training techniques, they often struggle with out-of-distribution generalization when deployed in novel, high-risk traffic scenarios. Third, safe exploration remains a significant open challenge, as conventional RL exploration strategies may lead to high-risk behaviors during training, potentially resulting in catastrophic failures. Safe RL techniques, such as conservative policy learning, shielding mechanisms, and risk-aware exploration strategies, require further refinement to balance exploration efficiency and safety constraints. Future research should focus on developing provably safe MARL frameworks, improving robustness to real-world uncertainties, and advancing risk-sensitive training methods that enable MARL agents to operate reliably in diverse, dynamic, and unpredictable traffic conditions.

B. Communication Challenges

1) Current Progress: Effective communication is pivotal in MARL for CAVs, enabling collaboration, information sharing, and coordinated decision-making through reliable protocols [172]. First, the development of advanced communication protocols tailored specifically for CAVs is a pressing need. Existing research focuses on protocols that enable efficient data sharing [173]. For instance, research endeavors have delved into the utilization of graph-based MARL algorithms [122], such as those explored in [136] and [138], to enable efficient and adaptive message exchange among agents. Second, the challenge of ensuring robust communication in dynamically evolving traffic scenarios represents another critical aspect that deserves meticulous attention [174]. For example, in [157], a variational auto-encoder algorithm combined with advanced multi-head attention mechanisms is proposed to extract and retain valuable information from neighboring agents while making efficient use of constrained communication resources. Last but not least, it is worth noting that managing unstable communication channels, particularly in terms of latency, emerges as a particularly intriguing consideration in the context of MARL for CAVs [108]. In [108], the double deep Q-learning algorithm is proposed to jointly train the agents to maximize the sum rate of V2N links while ensuring the desired packet delivery probability for each V2V link within specified latency constraints.

2) Research Gaps: While communication in MARL is a well-explored domain, its adaptation and application within CAVs present distinctive research gaps, which necessitate further exploration [132]. For instance, existing works often overlook pressing real-world concerns in the realm of CAVs, such as communication costs and the challenges posed by noisy environments. These critical factors, when ignored, can introduce substantial obstacles to the practical deployment of CAV technology. MARL algorithms applied in computer science domains are expected to be explored for use in CAV applications within challenging environments marked by limited bandwidth [59] and noisy communication channels [175]. Moreover, the majority of current MARL algorithms in CAVs rely on CTDE (Sec. II), potentially posing a significant privacy threat within the realm of communication. In the field of single-agent RL, security and privacy issues have garnered extensive attention [176]. However, in the context of MARL, and similarly in the domain of CAVs, these areas have remained relatively unexplored. This suggests a promising area for future research that merits exploration.

C. Mixed-Traffic Challenges

1) Current Progress: One of the formidable challenges posed by CAVs stems from navigating the intricate dynamics of mixed traffic, wherein CAVs share the road with various other road users, including human-driven vehicles, bicycles, e-scooters, and an array of other modes of transportation [177], [178]. This multifaceted coexistence necessitates advanced MARL algorithms to ensure seamless and safe interactions among these diverse road users. Current research into MARL algorithms for CAV applications primarily concentrates on integrating the prediction of human behavior into the decision-making process [179]. For instance, in [30], the authors introduce a novel MARL framework with a priority-based safety supervisor designed to preempt potential collisions. This is achieved by predicting future trajectories, encompassing both CAVs and HDVs and then the unsafe actions generated by MARL agents will be replaced with safe actions, thereby contributing to the enhancement of safety in mixed traffic scenarios. However, it is important to note that human-driven vehicles can introduce an additional layer of unpredictability into the traffic environment, since mixed traffic involves a diverse range of road users, presenting unique challenges for the seamless integration of CAVs into the existing road ecosystem.

2) Research Gaps: Given the diverse spectrum of road users, future research endeavors will encompass the development of real-time, adaptive safety assessment and intervention strategies capable of responding to the dynamic nature of traffic conditions. Another significant aspect entails the establishment of novel coordination and communication mechanisms between CAVs and other road users. For instance, as the presence of e-scooters on the roads continues to surge, there is a growing imperative to develop efficient methods for coordinating the interactions between CAVs and e-scooters [180]. Furthermore, the existing MARL algorithms are typically trained within specific scenarios, underscoring the necessity for research aimed at enhancing their capacity to generalize across a wide range of mixed traffic conditions and environmental contexts.

D. Sim-to-Real Challenges

1) Current Progress: The transition from simulated training environments to real-world deployment remains a fundamental challenge in MARL-based CAV control. While deep MARL has achieved significant breakthroughs in simulation environments (see Sec. III-D), models often suffer from performance degradation in real-world traffic due to discrepancies in traffic dynamics, sensor uncertainties, and unforeseen driving behaviors. To mitigate this sim-to-real gap, researchers have explored domain adaptation techniques, risk-sensitive training frameworks, and hybrid offline-online RL approaches. Offline RL, which trains policies using pre-collected datasets, has

also been considered to improve safety and data efficiency. However, its applicability to MARL-based CAVs remains an open challenge. Another emerging approach for enhancing generalization and policy transfer is Transformer RL (TRL). Unlike traditional RL architectures that rely on recurrent networks like LSTMs, TRL leverages self-attention mechanisms to capture long-range dependencies in sequential decision-making. This enables MARL agents to better process complex traffic patterns, adapt to diverse driving scenarios, and improve multi-agent coordination. Moreover, TRL scales efficiently with large interaction datasets, making it a promising tool for handling the high-dimensional decision-making required for real-world CAV deployment.

2) Research Gaps: The primary challenge in sim-to-real transfer lies in the inherent unpredictability of real-world environments, where traffic behaviors, adversarial interactions, and rare safety-critical events deviate from synthetic training data [181]. Most MARL policies struggle to generalize beyond controlled simulation settings, as existing domain adaptation techniques fail to account for complex, evolving dynamics. Real-world uncertainties, including sensor noise, connectivity disruptions, and multi-agent coordination failures, further degrade performance. Additionally, the lack of real-world benchmarks and standardized testing frameworks hinders objective evaluation, making it difficult to measure and compare progress in sim-to-real adaptation [182]. Ethical, legal, and regulatory constraints further complicate direct deployment, as autonomous decision-making in safety-critical scenarios demands a high level of explainability, accountability, and formal safety guarantees. Future research must focus on developing adaptive, risk-aware policies, leveraging real-time learning frameworks, improving uncertainty quantification, and establishing regulatory-compliant evaluation methodologies to ensure the scalable and responsible deployment of MARL-based CAV control.

V. Conclusion

Recently, MARL has emerged as a focal point of interest within the realm of CAVs. The capability of MARL to tackle intricate control and coordination challenges in CAVs has unlocked new horizons for the development of intelligent and efficient next-generation transportation systems. This review has undertaken a comprehensive exploration of the applications of MARL in the control of CAVs.

1) The review began with an overview of single-agent RL techniques and an extensive survey of the diverse landscape of MARL architectural variants, facilitating a deeper understanding of applications to both individual and collective agent behaviors.
2) Our analysis categorized these contributions based on different degrees of control, ranging from one-dimensional to two-dimensional and three-dimensional collaboration. Within this multifaceted arena, MARL has exhibited remarkable prowess, demonstrating its promising performance in various CAV control tasks. These include cooperative endeavors such as platooning control, lane-changing maneuvers, on-ramp merging, traffic signal coordination, and unsignalized intersections, among others.
3) We highlighted significant challenges in designing and evaluating MARL methods within the CAV. Striking a balance between performance, scalability, and safety remains a complex challenge.

This review serves as a crucial resource, aimed at encouraging further research and the practical application of MARL within the field of CAVs.

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